

# "What Can I Cook?" LetMeCook: An LLM-Based Interactive System for Personalized Recipe Generation

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## Abstract

In real-world cooking scenarios, users often need to create personalized recipes based on limited ingredients, dietary goals, and various restrictions, such as the availability of equipment, flavor preferences, and health conditions. Existing recipe generation methods lack the flexibility and adaptability to meet these individual needs. We present LetMeCook, an end-to-end interactive system for personalized recipe generation that leverages multimodal perception, hybrid retrieval, and content generation. Given a photo of the user's refrigerator and a dietary profile, LetMeCook detects available ingredients, retrieves relevant recipe candidates, and generatively refines them based on user requirements, such as ingredient substitution and flavor adjustment. The system provides both textual and visual previews of the adapted recipes, offering a highly interactive and user-centric experience.

## CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Computer vision**; • **Information systems** → **Information systems applications**; • **Human-centered computing** → **Interaction design**.

## Keywords

Personalized Recipe Generation; Large Language Model; Multimodal Interaction

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## 1 Introduction

In everyday life, users often want to discover recipes that they can cook using mainly the currently available ingredients. Although previous studies [1, 2, 6, 9, 11, 13] propose various methods for

generating recipes based on given ingredients, these approaches fail to accommodate the highly personalized and constraint-rich cooking scenarios encountered in real life. On the one hand, users may not have all the ingredients required for a standard recipe. On the other hand, they may face additional constraints related to kitchen equipment, available seasonings, individual flavor preferences (e.g., mild or spicy), nutritional goals (e.g., muscle gain, sugar control), and health restrictions (e.g., diabetes, food allergies). These factors make it challenging to generate feasible and personalized recipes. Furthermore, even after making an initial recipe recommendation, users may wish to refine the recipe further according to their needs, highlighting the importance of dynamic recipe adaptation in interactive systems.

To address these challenges, we introduce **LetMeCook**, an end-to-end personalized recipe generation system that integrates multimodal perception, hybrid retrieval, and content generation. Given a refrigerator photo and the user's dietary profile, the system first detects the available ingredients in the photo. Then, it incorporates the user's dietary profile to query a recipe dataset and retrieve the most suitable candidate recipes. We support interactive and generative adaptations to the retrieved recipes based on user requirements, including substituting missing ingredients, flavor customization, and nutritional adjustment. All adaptations are accompanied by natural language explanations. Finally, users can preview both the textual and visual versions of the modified recipe, enabling a fully interactive experience.

## 2 System Architecture

Figure 1 illustrates the overall architecture of our LetMeCook system, which is composed of five main modules: (1) Ingredient Detection, (2) Hybrid Recipe Retrieval, (3) Candidate Selection, (4) Recipe Adaptation, and (5) Image Regeneration. These modules are designed to integrate zero-shot visual recognition, hybrid retrieval, textual content generation, and visual content generation in a seamless pipeline that supports interactive and personalized recipe generation.

**Ingredient Detection.** Given a photo of their refrigerator, we perform ingredient recognition using a three-stage, zero-shot vision pipeline: (i) First, we apply YOLOv8 Nano [12] to generate bounding boxes, which we treat as region proposals for visible objects. Unlike traditional classifiers, we discard the COCO labels and treat YOLO as a generic object detector. (ii) Each detected region is cropped and passed to the CLIP [7] model for the classification of the open-vocabulary ingredients. We use labels consisting of ingredient names from Recipe1M [5], augmented with a predefined

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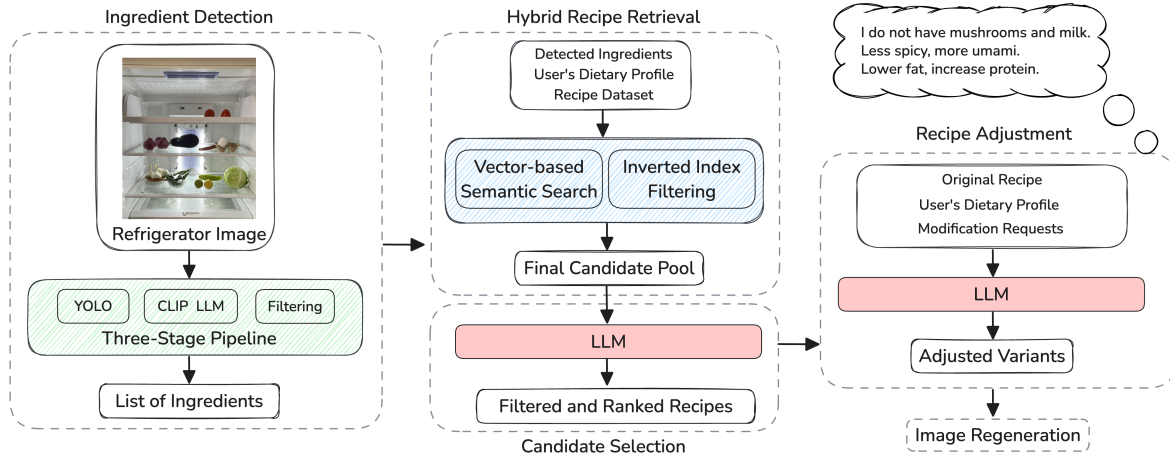


Figure 1: System architecture of LetMeCook.

set of common container types (e.g., "jar," "bottle," "box," "can") to help detect packaged food items. CLIP computes similarity scores between each region and all candidate labels, and we retain the top-1 prediction for each region. Regions with low confidence (score  $< 0.25$ ) and those classified as containers are passed to LLM (e.g., GPT-4o-mini [3]) for further reasoning, leveraging OCR outputs and visual context to infer the specific food item. (iii) The ingredient list is filtered to remove any allergens or disliked items based on the user's profile. Finally, the list of ingredients will be presented to the user for confirmation.

**Recipe Retrieval.** After confirming the available ingredients, we retrieve the candidate recipes using a hybrid approach that combines vector-based semantic search and inverted index filtering, because dense embeddings capture latent dietary intent and stylistic similarity but may overlook exact ingredient constraints, whereas the inverted index retrieves recipes containing the user's key ingredients but ignores the semantic relation. Fusing both outputs yields a pool that is semantically and ingredient relevant, giving the downstream LLM reranker richer complementary signals for selecting the final recipes. For vector-based retrieval, we first build the vector database offline. Specifically, we concatenate each recipe title, description, and ingredient list from the recipe dataset into a single text string, and embed it using Sentence-BERT [8]. The resulting embeddings are indexed using Faiss [4] for an efficient approximate nearest neighbor (ANN) search. At query time, the user's dietary profile (concluded by gpt-3.5-turbo) and detected ingredients are similarly concatenated and embedded to perform top-10 ANN retrieval. For the inverted index, we compute the IDF [10] scores for all ingredients in recipes. During retrieval, we select the top-10 refrigerator ingredients ranked by IDF scores, retrieve all recipes containing at least one of them, and compute a coverage score for each recipe:  $\text{Coverage}(R) = \frac{|F \cap R|}{|R|}$ , where  $F$  denotes the set of ingredients detected from the refrigerator, and  $R$  is the set of ingredients required for a given recipe. Then, the top-10 recipes ranked by coverage score are combined with the vector search results to form the final candidate pool with 20 candidates.

**Candidate Selection.** The retrieved recipes are passed to LLM (e.g., GPT-4o-mini) to select the top-5 candidates and re-rank the left

candidates. Recipes are filtered and ranked according to aligning with the user's dietary profile and minimizing missing ingredients. Each recommended recipe is accompanied by an explanation (e.g., "uses eggs, avoids dairy") and a list of missing ingredients.

**Recipe Adjustment.** Inspired by the TextGrad [14, 15] algorithm, we treat the LLM as an optimizer to adjust recipes based on different user requirements. Users can request modifications in three categories: missing ingredients (e.g., "I do not have mushrooms and milk."), flavor customization (e.g., "Less spicy, more umami."), and nutritional adjustment (e.g., "Lower fat, increase protein."). We construct the natural language prompt using the original recipe (title, description, ingredients), the user's dietary profile, and the specific modification requests. The LLM-based optimizer (e.g., GPT-4o-mini) then generates three adjusted variants, where newly introduced ingredients are annotated with <add> tags, and each change is explained using <reason> tags. All variants are validated to exclude allergens, and users can continue to interact with the system until a satisfactory version is obtained.

**Image Regeneration.** When the recipe is modified, we regenerate the image of the dish using GPT-Image-1<sup>1</sup> to help users better visualize the updated recipe. A natural language prompt is constructed based on the new recipe's title and ingredients, and the original image is used as a visual reference. Then the GPT-Image-1 API generates a new image that reflects the modified dish. To ensure visual-textual consistency, an LLM (e.g., GPT-4o-mini) is employed to validate whether the generated image aligns with the modified textual content. If validation passes, the image is presented to the user; otherwise, the regeneration is retried.

### 3 Conclusion

We present LetMeCook, a multimodal and interactive system that enables personalized recipe generation through ingredient detection, recipe retrieval, adaptive adjustment, and image regeneration. From what is in your fridge to what is on your plate, LetMeCook makes everyday cooking more feasible and user-friendly.

<sup>1</sup><https://platform.openai.com/docs/guides/image-generation?image-generation-model=gpt-image-1>

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